

Ahmed Taleb Bechir

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PROFILE

Seeking an opportunity in full-stack development, cloud/DevOps engineering, or QA engineering. Software engineer with hands-on experience in backend systems, cloud deployments, and test automation. ENS Paris-Saclay Master's graduate with a strong C/C++ and systems foundation, now building web services, CI/CD pipelines, and end-to-end testing workflows.

EDUCATION

École Normale Supérieure Paris-Saclay <i>Master M2 – Modeling and Simulation for High-Performance Computing</i>	2024 – 2025
Université Paris-Saclay <i>Master M1 – High-Performance Computing & Simulation (CHPS)</i>	2023 – 2024
University of Strasbourg <i>Bachelor's Degree in Computer Science</i>	2019 – 2023

PROFESSIONAL EXPERIENCE

R&D HPC Engineering Intern, Eviden (Bull) March 2025 – September 2025 Bruyères-le-Châtel, France

- Designed and implemented an open-source transport layer in C, validated on two distinct cluster interconnects (InfiniBand and BXI), ensuring cross-platform portability. Built multi-library toolchains across heterogeneous targets.
- Worked in an agile R&D team, contributed to client-facing technical demonstrations; focus on low latency and scalability.

IT Observation Intern, Banque Populaire de Mauritanie June 2023 – August 2023 Nouakchott, Mauritania

- Contributed to the migration of legacy infrastructure to VMware vSphere as part of a bank-wide digital transformation.
- Supported data validation and operational reporting during the deployment of a nationwide digital payment system.

PROJECTS

Fleet Management OS	Personal project – 2025
- Built a full stack fleet management platform with TypeScript end-to-end: Express/Node.js REST API with Zod validation, React frontend with React Query for server state management, MongoDB Atlas with Mongoose ODM. - Implemented domain logic including vehicle status state machine with enforced transitions and maintenance scheduling with interval overlap conflict detection (compound index on vehicleId, scheduledStart, scheduledEnd). - Deployed backend on Google Cloud Run via multi-stage Docker build, frontend on GitHub Pages via GitHub Actions CI/CD. Added Kubernetes manifests with liveness and readiness health probes.	

Multiplayer Web Game (Saboteur Online)	Team of 11
- Built a real-time multiplayer card game: REST API for lobby/auth/matchmaking, WebSocket layer (Socket.IO) for in-game state with hidden-information partitioning, MongoDB persistence. Owned backend API, database design, and networking. - Containerized with Docker, load-tested with Artillery, API documented with OpenAPI/Swagger.	

RAG Application on GCP Cloud Run	Personal project
Built a retrieval-augmented generation service and deployed it on GCP Cloud Run with a Docker-based CI/CD pipeline via Cloud Build.	

Black-Scholes Monte Carlo Optimization, Teratec Hackathon 2025	2025
Achieved a 1000x speedup (904s to 0.87s) on an option pricer in C++ on ARM Graviton 4 via OpenMP parallelization and SIMD vectorization with NEON intrinsics. Profiled with Linux perf and flamegraphs.	

TECHNICAL SKILLS

Programming: C, C++, TypeScript, Python, Bash

Web/Backend: Node.js, Express, React, React Query, REST APIs, MongoDB, Mongoose, Zod, Socket.IO, FastAPI

Cloud/DevOps: GCP (Cloud Run, Cloud Build, Compute Engine), Docker, Kubernetes, Pulumi (learning), GitHub Actions

Testing/QA: Playwright, Artillery (load testing), Jest, structured test planning

HPC/Systems: MPI, OpenMP, CUDA, libfabric, SIMD/NEON

Tools: Git, CMake, Linux, GDB, Valgrind, Perf, Flamegraphs

Languages: French (native), English (C2), Arabic (C2)